Final Project Proposal: Sequential State-of-Charge Management for Battery Frequency Response Under Uncertainty

Victoria Dinov vdinov@stanford.edu
Lynne Irvin lynnei@stanford.edu

1 Goal

Grid operators keep system frequency close to its nominal value by procuring fast-acting reserves that inject or absorb power whenever frequency deviates. Battery energy storage systems are well suited to this role because they can respond within seconds, but they have a limitation that conventional generators do not: a finite energy capacity. A battery that has drifted to a low state of charge (SoC) cannot respond to an under-frequency event, and one that is nearly full cannot respond to an over-frequency event (Oudalov et al., 2007). Keeping the battery "ready" therefore requires actively managing its SoC between events, without knowing when the next deviation will occur, in which direction, or how large it will be.

Our goal is to formulate battery SoC management for frequency response as a Markov decision process (MDP), solve it with methods from the course (Kochenderfer et al., 2022), and compare the resulting policy against the simple rule-based strategies commonly used in practice. We will build the frequency model from real grid data (frequency, voltage, load, and historical contingency events) from an industry partner. Success means a policy that measurably reduces response shortfall relative to baselines at equal or lower battery cycling, together with a clear characterization of the trade-off between reliability and battery wear.

2 Decision Making

At each time step t (on the order of minutes), the battery must deliver a mandatory droop response $r_t = -K\,\Delta f_t$, clipped to its committed capacity, where Δf_t is the frequency deviation and K is the droop gain. On top of this, the controller chooses a small SoC-management power u_t that slowly charges or discharges the battery to restore headroom. The decision is sequential because every choice of u_t changes the SoC available for all future responses: charging now protects against a future under-frequency event but leaves less room for an over-frequency event, and every unit of energy moved adds degradation. We sketch the MDP as follows.

- **States** $s_t = (e_t, f_t)$: discretized SoC $e_t \in [E_{\min}, E_{\max}]$ and a discretized frequency-deviation bin f_t (e.g., deadband, small/large under-frequency, small/large over-frequency).
- **Actions** $u_t \in \{-u_{\max}, \ldots, 0, \ldots, u_{\max}\}$: a discrete set of SoC-management power levels.
- **Transitions**: SoC evolves as $e_{t+1} = \mathrm{clip}\big(e_t + \Delta t\,[\eta_c (p_t)^+ - (p_t)^-/\eta_d]\big)$, where $p_t = r_t + u_t$ is the net charging power, $(\cdot)^+$ and $(\cdot)^-$ denote its charging and discharging parts, and η_c, η_d are charge and discharge efficiencies. The frequency bin evolves stochastically according to a transition model $T(f_{t+1} \mid f_t)$ learned from data.
- **Rewards**: $R(s_t, u_t) = -\lambda_s\,\mathrm{shortfall}_t - \lambda_d\,|p_t|\,\Delta t - \lambda_e\,(u_t)^+ \Delta t$, penalizing response the battery could not deliver because of SoC limits, energy throughput as a proxy for degradation (Xu et al., 2018), and energy purchased for SoC management. We use a flat energy cost λ_e.

We will use an infinite-horizon discounted formulation.

3 Sources of Uncertainty

The dominant source of uncertainty is the frequency deviation process: its direction, magnitude, and persistence are not known in advance. We will model it as a Markov chain over deviation bins and estimate $T(f' \mid f)$ by maximum likelihood with a Dirichlet prior, since large-deviation bins are rare and raw counts would be unreliable. The second source is contingency events (e.g., sudden generator or line trips), which are rare but cause the largest, most sustained deviations. We will use the historical event log to estimate their arrival rate and typical profile, and splice realistic events into evaluation traces to stress-test policies.

A third source is model uncertainty: real frequency is likely not exactly Markov at our chosen resolution, and battery parameters (efficiency, degradation cost) are only approximately known. We will address this by evaluating all policies on held-out historical frequency traces rather than only on the learned model, and by sensitivity analysis over the battery parameters and penalty weights. Finally, an unobserved grid regime (normal vs. stressed) could add partial observability (see below).

4 Sketches of Solution

With roughly 21 SoC levels, 5–11 frequency bins, and 5–9 actions, the state space has only a few hundred states, so we can solve the MDP exactly with value iteration and extract the greedy policy. We will compare it against three baselines: (1) pure droop response with no SoC management, (2) a proportional controller that steers SoC toward a fixed target (e.g., 50%), a common industry heuristic, and (3) a myopic policy that minimizes only the immediate expected shortfall. Metrics will include the fraction of intervals with full response delivered, total shortfall energy, equivalent full cycles, and the time spent near SoC limits. Sweeping λ_s/λ_d will trace out a reliability-versus-wear trade-off curve for each policy.

An added goal would be for us to add a hidden grid regime $z_t \in \{\text{normal}, \text{stressed}\}$ that changes the frequency dynamics. The controller would observe only measured frequency and voltage, maintain a belief over z_t using a hidden Markov model fit to the data, and act on that belief (e.g., via QMDP). This quantifies the value of using grid-stress signals.

AI Disclosure

We used Claude to scope project ideas, help formulate the MDP, and debug LaTeX formatting.

References

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